Task 4: PHP Website Hosting in User Home Directory

Objective

1. Create a new user named Alice.
2. Create a sample web page under a directory named public_html: /home/Alice/public_html
3. Access the website using the following URLs:
   myfirstwebsite.test
   www.myfirstwebsite.test

Webpage content:

```
<?php
phpinfo();
?>
```

Requirements

- AWS EC2 instance running RHEL
- Apache (httpd) web server
- PHP
- User account named zenhal
- public_html directory
- PHP webpage using phpinfo()
- Custom Apache Virtual Host configuration

Procedure

Step 1 – Create the User

A new user named **zenhal** was created on the RHEL system.

sudo useradd zenhal

Step 2 – Create the public_html Directory

A public_html directory was created inside the user's home directory.

sudo mkdir -p /home/zenhal/public_html

Step 3 – Create the PHP Webpage

A PHP webpage named index.php was created inside the public_html directory.

sudo vim /home/zenhal/public_html/index.php

The following PHP content was added:

```
<?php
phpinfo();
?>
```

Step 4 – Create the Custom Virtual Host Configuration

A custom Apache Virtual Host configuration file was created for the website.

sudo vim /etc/httpd/conf.d/myfirstwebsite.conf

The following configuration was added:

```
<VirtualHost *:80>
   ServerName myfirstwebsite.test
   ServerAlias www.myfirstwebsite.test

   DocumentRoot /home/zenhal/public_html

   <Directory /home/zenhal/public_html>
      Require all granted
   </Directory>

   DirectoryIndex index.php
</VirtualHost>
```

Step 5 – Test the Apache Configuration

The Apache configuration was tested to check for syntax errors.

sudo apachectl configtest

The result was:

Syntax OK

Step 6 – Restart Apache

Apache was restarted to apply the new configuration.

sudo systemctl restart httpd

Step 7 – Configure Directory Permissions

The required permissions were configured for the user's home directory and web directory.

sudo chmod 711 /home/zenhal

sudo chmod 755 /home/zenhal/public_html

Step 8 – Check SELinux

The SELinux status was checked.

getenforce

The result was:

Enforcing

Step 9 – Allow Apache to Access User Home Directories

Apache was allowed to access web content located inside user home directories.

sudo setsebool -P httpd_enable_homedirs on

The public_html directory retained the appropriate httpd_user_content_t SELinux context.

Step 10 – Install PHP

PHP was installed on the RHEL system.

sudo dnf install php -y

The PHP installation was verified using:

php -v

Step 11 – Verify Apache and Port 80

The Apache service was checked to confirm that it was active and running.

sudo systemctl status httpd

Port 80 was checked to confirm that Apache was listening for HTTP connections.

sudo ss -lntp | grep :80

Step 12 – Test the Custom Virtual Host

The custom Virtual Host was tested using the required hostname.

curl -I -H "Host: myfirstwebsite.test" http://127.0.0.1/

The result was:

HTTP/1.1 200 OK

This confirmed that Apache was correctly serving the custom Virtual Host.

Step 13 – Verify the PHP Website

The PHP webpage was tested through the custom Virtual Host.

curl -H "Host: myfirstwebsite.test" http://127.0.0.1/

The PHP information page was returned successfully, confirming that the phpinfo() page was executing correctly.

Step 14 – Access the Website Using a Web Browser

The hosted PHP website was accessed through a web browser using the configured domain name.

The following URLs were used:

http://myfirstwebsite.test

and

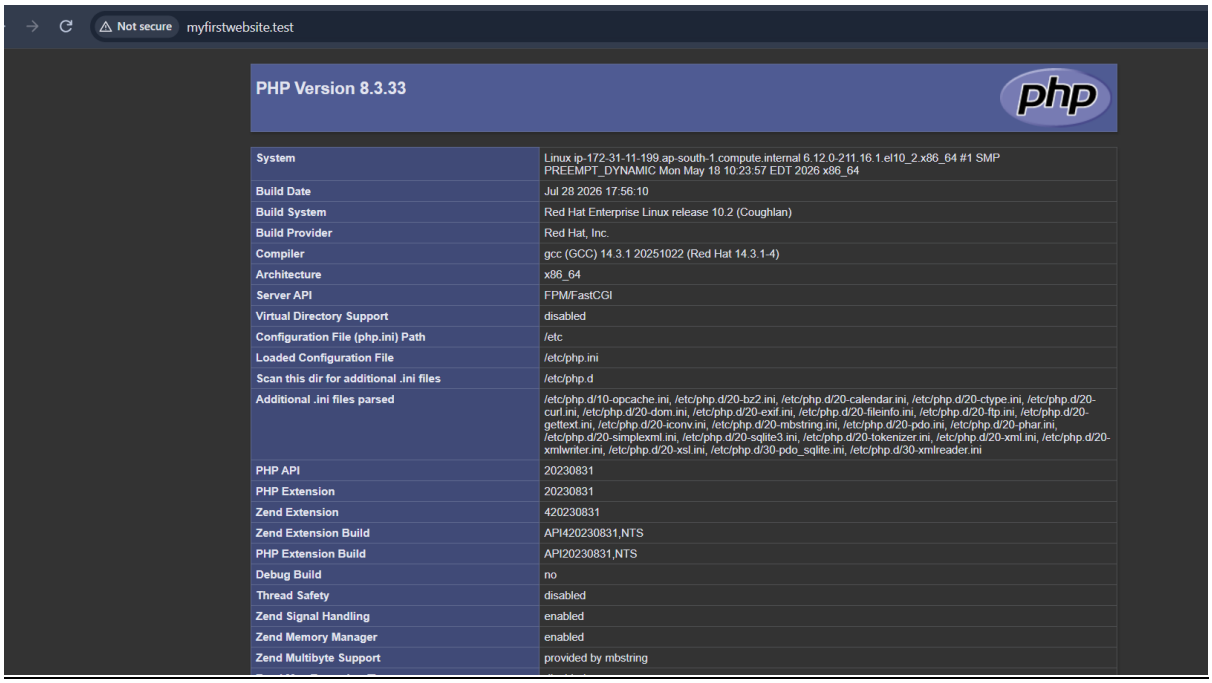
http://www.myfirstwebsite.test

The PHP information page generated by phpinfo() was displayed in the web browser, confirming that the website was successfully hosted and accessible through the configured domain names.

Using:

http://myfirstwebsite.test

http://www.myfirstwebsite.test



Verification

- The user **zenhal** was successfully created.
- The /home/zenhal/public_html directory was successfully created.
- The index.php file was created with phpinfo().
- Apache Virtual Hosting was configured for myfirstwebsite.test.
- www.myfirstwebsite.test was configured as a ServerAlias.
- Apache configuration returned **Syntax OK**.
- Apache was confirmed active and running.
- Apache was confirmed to be listening on port 80.
- SELinux was running in **Enforcing** mode.
- Apache was allowed to access the user's home directory.
- The custom Virtual Host returned **HTTP 200 OK**.
- The PHP phpinfo() page was successfully executed.

Conclusion

Through this task, a simple PHP website was successfully hosted inside the home directory of the user **zenhal** on a RHEL AWS EC2 instance. Apache Virtual Hosting was configured for **myfirstwebsite.test** and **www.myfirstwebsite.test**. The required directory permissions and SELinux settings were configured to allow Apache to access the user's web directory. PHP was installed and verified, and the phpinfo() webpage was successfully executed, confirming that the PHP website was working correctly.